Data Scientist Roadmap

A structured roadmap for becoming a Data Scientist.

1. Mathematics & Statistics

Learn probability, statistics, linear algebra, and calculus. Key topics: mean, median, variance, hypothesis testing, distributions, matrices, vectors, eigenvalues, gradients, and optimization.

2. Programming Fundamentals

Master Python. Learn variables, loops, functions, OOP, file handling, exception handling, and package management.

3. Data Analysis

Learn NumPy, Pandas, and data cleaning techniques. Work with missing values, feature engineering, and exploratory data analysis (EDA).

4. Data Visualization

Use Matplotlib, Plotly, and dashboards. Learn to communicate insights through charts and reports.

5. SQL & Databases

Learn SQL queries, joins, subqueries, indexing, normalization, and relational database concepts.

6. Machine Learning

Study supervised and unsupervised learning. Learn regression, classification, clustering, decision trees, random forests, SVMs, and model evaluation metrics.

7. Deep Learning

Learn neural networks, CNNs, RNNs, LSTMs, Transformers, TensorFlow, PyTorch, and modern generative AI concepts.

8. Big Data & Cloud

Learn Spark, Hadoop basics, cloud platforms (AWS, Azure, GCP), and scalable data processing pipelines.

9. Projects

Build EDA projects, predictive models, recommendation systems, NLP applications, and end-to-end data science portfolios.

10. MLOps & Deployment

Learn Flask/FastAPI, Docker, Git, CI/CD, monitoring, and model deployment techniques.

Suggested Timeline

Months 1-2: Python, Statistics, SQL
Months 3-4: Data Analysis and Visualization
Months 5-6: Machine Learning
Months 7-8: Deep Learning
Months 9-10: Cloud, Big Data, MLOps
Months 11-12: Portfolio Projects and Interview Preparation